

the same affective content within the prompt does not systematically change LLM performance. By contrast, the paraphrased condition is consistently worse across all four models and nearly all emotions, often producing the largest drop relative to the original baseline. This pattern suggests that positional placement matters little, whereas paraphrasing the question introduces a larger distribution shift that is clearly more harmful for mathematical reasoning.

Model	Condition	Baseline	Prepended	Mid-Pos	Appended	Paraphrased
DeepSeek	Baseline (Original)	94.97	–	–	–	–
	Fear		95.07	94.47	94.47	92.80
	Anger		94.77	94.31	94.54	93.40
	Happy		95.12	94.67	94.52	93.47
	Sad		95.42	95.20	94.07	92.12
	Disgust		95.15	94.62	94.32	92.27
	Surprise		94.62	94.84	94.18	88.36
GPT-OSS	Baseline (Original)	94.67	–	–	–	–
	Fear		93.33	93.93	93.71	90.75
	Anger		92.87	93.40	93.78	92.19
	Happy		94.90	94.30	94.07	91.97
	Sad		93.77	93.47	93.47	90.99
	Disgust		94.32	94.02	93.56	91.52
	Surprise		93.07	94.40	93.66	86.96
Llama3.3	Baseline (Original)	95.00	–	–	–	–
	Fear		94.69	94.62	95.30	92.12
	Anger		94.62	94.47	94.62	92.57
	Happy		95.12	94.82	94.67	93.32
	Sad		94.82	94.82	95.05	91.82
	Disgust		95.00	94.70	94.92	91.67
	Surprise		95.06	94.62	94.99	87.18
Qwen3	Baseline (Original)	93.93	–	–	–	–
	Fear		94.01	93.63	93.25	89.84
	Anger		93.78	94.16	93.48	92.27
	Happy		94.15	94.60	94.22	92.65
	Sad		94.14	94.07	93.84	91.29
	Disgust		93.86	94.09	94.32	90.45
	Surprise		93.74	94.47	93.74	86.66

Table D2 Accuracy on GSM8K under different structural variants of emotional framing. For each model and emotion, we compare the original no-emotion baseline with three positional insertions of the same emotional cue (prepended, mid-position, and appended) and a paraphrased variant of the prompt. Across models, changing the position of the emotional cue produces only small differences, whereas paraphrasing the original question is consistently more harmful and yields the largest drops in accuracy.

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